Professor Mahendra P. Lama, senior advisor at the International Centre for Integrated Mountain Development and a former member of India's National Security Advisory Board, argues that Himalayan borders must be treated not only as security lines but as shared ecological spaces. "People, pilgrims, tourists, traders, rivers and natural hazards move" through these borderlands, he says, while the exchange of hydrological data remains thin even between India and Nepal and far more restricted with China. For the Indus, Sutlej and Brahmaputra systems, that is not merely an environmental problem, it is a national security vulnerability.

India needs automatic, real-time exchange of upstream river levels, rainfall, seismic signals, satellite detections and barrier lake alerts with Nepal, Bhutan and China, backed by joint protocols that operate even when political relations deteriorate. Data should flow directly to basin control rooms and district authorities, not wait for slow diplomatic clearance. Neighbouring countries also need common hazard maps, compatible alert thresholds and regular cross-border exercises. A flood respects gravity, not sovereignty.



"Until governments treat the Himalayas as a climate frontline rather than as a postcard backdrop, Rasuwa won't be the last name we're forced to memorise"

ANJAL PRAKASH
Author, IPCC

WARNING WITHOUT READINESS

India has not ignored the danger. After the Sikkim disaster, the Union government launched the Rs 150-crore National GLOF Risk Mitigation Programme for Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh. Its mandate includes hazard assessment, monitoring, early warning systems, structural measures and community preparedness. GLOF studies are now compulsory for proposed dams with vulnerable lakes in their catchments, and design assumptions are being reviewed for exposed existing and under-construction projects.

But the gap between seeing danger from a satellite and moving a family to safety remains wide. The programme envisaged early-warning systems at 195 high-risk lakes. Officials say deployment at the lakes has fallen behind the original timetable, as funds arrived late, equipment is still being tested and expeditions to remote sites are possible only in short weather windows. Krishna Vatsa, head of department at NDMA (National Disaster Management Authority), says systems are in advanced testing and that some installations in Himachal Pradesh are imminent. However, as he points out, "Simply installing is not enough; they need to be able to communicate," recalling how one system stopped working and was difficult to inspect.

Experts say a credible end-to-end warning network is more than a water gauge beside a lake. Sensors at the source must measure water level, river discharge, rainfall, ground vibration and sediment. Seismic instruments should detect a large avalanche the moment it begins. The information must travel through duplicate satellite, radio and telecom links to a staffed control room, hydro operators, district officials and communities downstream. Pre-set thresholds should automatically trigger sirens, loudspeakers, mobile alerts, road closures, power-project shutdowns and evacuation. Everyone must know who has the authority to act and where people will go.

This is vital because the warning window may be measured in minutes. A phone tower, gauge or bridge in the flood path may be the first thing destroyed. Nepal's system did issue alerts and save lives, Haritashya notes, but it was not designed for a debris flow of this magnitude. "We need to put early warning systems much further upstream, so that there is enough time for people downstream to react, with a clear chain of command established beforehand—not figured out in the moment."

Nor can early warning be the only defence. At selected accessible lakes, engineers can lower the water level through siphons, pumps or controlled outlet channels, reducing the volume that makes for a catastrophic release. Nepal did this at Tsho Rolpa and Imja. Such measures are difficult and expensive in India, and they address only known lakes—not a sudden bedrock and glacier collapse. They should therefore be reserved for the lakes where downstream lives and critical infrastructure justify intervention, while catchment-wide monitoring covers the hazards that cannot be drained.

A siren cannot also protect a dam or compensate for reckless construction. "No siren can outrun a mountainside collapsing in seconds," warns Anjal Prakash, professor of public policy at FLAME University and an author at IPCC (Inter Governmental Panel on Climate Change). The first line of defence is to keep new settlements, hotels, fuel depots, worker colonies and critical facilities out of mapped inundation and debris-flow corridors, enforce river setbacks, control blasting and the dumping of construction muck, and provide multiple evacuation routes to high ground. Existing high-risk sites need retrofitting, drills and, where necessary, relocation.

WHAT INDIA MUST DO NOW

The Nepal disaster changes the question India must ask. It is no longer enough to identify which moraine-dammed lake might burst. Planners must ask which entire mountainsides can fail, what they can strike, where debris could block a river, how fast successive surges would travel, and what lies in their path. Experts say that among the first things India must do is replace the static lake list with a con-